

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Glucose → carbon dioxide + ethanol Energy neutral Yeast above arrow neutral	1			1		
	(b)	(i)		{Transfer/exchange/movement} of <u>gene</u> from one organism to another	1			1		
		(ii)		The maximum of yeast is at 4.5 (days) (1) But on this day the sugar content was 60 (units) {and still falling/which is not the minimum} (1) And ethanol concentration was 2.8 (%) {and still rising / which is not the maximum} (1) (so disagree) no data max (1) e.g. when yeast is at a maximum, sugar content {still falling/not at a minimum} and ethanol content {still rising/not at a maximum}			3	3	3	
	(c)			Any 2 ×(1) from: (Long term human) health effects are unknown (1) Effects on insects unknown (1) Any correct statement about resistance (1) May cross pollinate with other plants/yeast (to produce super yeast that would be difficult to kill) (1) Out-compete other plant species (1) Reduce gene pool (1)	2			2		

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	(d)			Any 4 × (1) from: Mixture is boiled (1) Mixture is cooled (1) yeast is added (1) Allow fermentation (1) Filter (1) Must be in sequence For 4 marks yeast must be added	4			4		
				Question 4 total	8		3	11	3	